Sofia Ramos

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EDUCATION

Olin College of Engineering | Needham, MA
Bachelor of Science in Mechanical Engineering | Expected May 2029 | GPA: 4.0
Relevant coursework: Mechanics of Solids and Structures, Mechanical Prototyping, Principles of Integrated Engineering

ENGINEERING EXPERIENCE AND PROJECTS

Prosthetic Hands | High School Capstone | Spring 2025

- Designed, built, and tested three Arduino-controlled, DC-motor-driven prosthetic hand and wrist assemblies, each validated with a custom acrylic fixture for wrist rotation, flexion, or deviation

Formula SAE Cockpit Engineer | Olin Electric Motorsports | Fall 2025–Present

- Designed the head restraint and mounting tabs in SolidWorks to meet the Formula SAE 900 N load requirement for the team's 2027 competition car
- Verified the mounting tabs under a 900 N rearward load using hand calculations and von Mises stress FEA, achieving a factor of safety of 4
- Produced the engineering drawing and prepared the water jet cut file for the mounting tabs and plasma cut the aluminum head restraint

Wave Kinetic Sculpture | Course Project | Spring 2026

- Designed a motor-driven kinetic sculpture in SolidWorks, using shaft-mounted cams to turn rotary motion into vertical wave motion
- Laser cut the wave elements, cams, and spacers, then assembled and tested the mechanism with teammates, resolving friction issues with low-friction tape

Desk Companion | MIT Hackathon | Spring 2026 | 2nd Place, Healthcare Track

- Designed and iterated phone holders in Onshape for a posture and focus companion and integrated hardware with teammates' software for a live demo during a 48-hour hackathon

WORK EXPERIENCE

Laguna Blanca High School | Santa Barbara, CA | Assistant Counselor | Summers 2024–2026

- Supervised 20+ campers ages 4–13 each week and led hands-on science activities, including micropipetting and bacterial growth experiments

TECHNICAL SKILLS

- **Programming:** MATLAB, C++ (Arduino)
- **CAD and Analysis:** SolidWorks, Onshape, FEA, Hand Calculations
- **Design Documentation:** Engineering Drawings, Bill of Materials (BOM)
- **Fabrication:** 3D printing, mill, lathe, plasma cutting
- **Languages:** English (Native), Spanish (Advanced)